Version: 1.6

Bwrdd Iechyd Prifysgol Betsi Cadwaladr University Health Board

ANASTHETICS, CRITICAL CARE & PAIN MANAGEMENT CPG

Management of pre-operative anaemia in major adult elective surgery-WEST

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Documents to be read alongside this policy:	BCUHB Blood Transfusion Requirements
The aim of the document is to guide pre-operative healthcare professionals in screening patients for and optimising patients with anaemia who are to undergo major surgery thereby reducing transfusion requirements.	

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1. Introduction and Purpose of the Written Control Document

1.1	The aim of the document is to guide pre-operative healthcare professionals in screening patients for and optimising patients with anaemia who are to undergo major surgery thereby reducing transfusion requirements.
1.2	Anaemia is defined according to WHO criteria[1] as Haemoglobin (Hb) concentration <120g/L in women and <130g/L in men.
1.3	Pre-operative anaemia is common and can be found in as many as 75% of preoperative patients depending on the population studied and the type of surgery[3].
1.4	Pre-operative anaemia has been identified as independent risk factor for complications and death[4].
1.5	There is also evidence that transfusion of blood is associated with increased morbidity and mortality[5-7]. Reduction in the use of red cells following appropriate anaemia management is cost effective and conserves blood supplies for those patients who need it most.

2. Scope

2.1	This document applies to all adults undergoing major surgical procedures at Ysbyty Gwynedd.
2.2	A major surgical procedure is defined, in this context, as any procedure for which a group and screen or cross match is recommended. (See BCUHB Blood Transfusion Requirements document)
2.2	This document is intended for all physicians, nurses and pharmacists in the pre operative assessment clinic (POAC).

3. Content of Clinical Guideline

3.1. Responsibilities

3.1.1	POAC Nurse
	Identify patients who are undergoing major surgery.
	Check Hb from a blood test within 4 weeks. Otherwise perform 'Hemocue' test to identify patients with anaemia.
	Perform regular calibration tests in 'Hemocue' devices.
	Send additional 'anaemia' tests when appropriate.
	Review blood test results no later than the next working day.
	Liason with POAC pharmacist and/or doctor to agree on plan of care.
	Arrange appointment for intravenous therapy.
	Liaise with POAC pharmacist if any treatment is required (vitamin B12, folate, oral and intravenous iron, epo).
	Provide patient information and explain the benefits and risks of therapy offered.
	Safe administration of intravenous medications. Trained in evaluation and management of Anaphylaxis.
	Liaise with the primary healthcare team to inform them of any treatment commenced or planned.
	Arrange and review post therapy blood tests

3.1.2	POAC Pharmacist
	Prescription and dispensing of any anaemia therapy required: vitamin B12, folate, oral and intravenous iron.
	Provide patient information with regard to therapy offered.

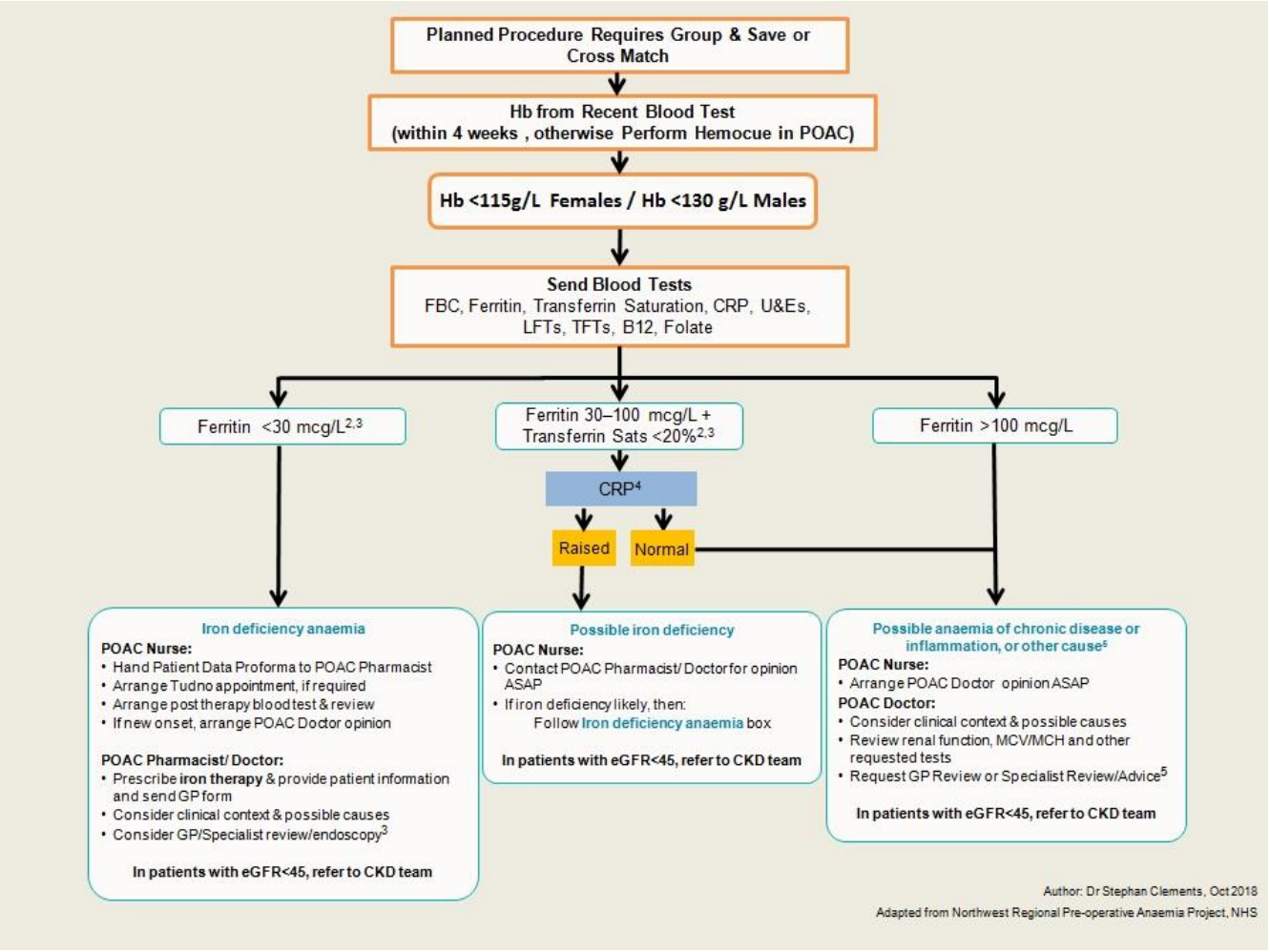
3.1.3	POAC Anaesthetist
	Review anaemia investigations. Consider clinical context i.e. cause, co-morbidities, likely peri-operative blood loss and agree on plan of care
	Prescribe intravenous therapies
	Liaise with haematology, gastrointestinal specialities and surgeon as appropriate

3.1.4	Primary care team
	To review investigations forwarded from POAC and act on them accordingly.

3.1.5	Patient/Carer
	Understand the need for investigation and treatment of anaemia
	Understand the benefits and risks of intravenous therapy
	To inform POAC if unable to tolerate oral iron.

3.2 . Guidance

3.2.1 Flow Chart of Guidance



3.2.2	Evaluation of Anaemia
	Anaemia may be multi-factorial, especially in the elderly or in those with chronic disease, renal impairment, nutritional deficiencies or malabsorption.
	In an anaemic adult, a ferritin level <15 mcg/L is diagnostic of iron deficiency, and levels between 15–30 mcg/L are highly suggestive. However, ferritin is elevated in inflammation, infection, liver disease and malignancy. This can result in misleadingly elevated ferritin levels in iron-deficient patients with coexisting systemic illness. In the elderly or in patients with inflammation, iron deficiency may still be present with ferritin values up to 60–100 mcg/L.
	Patients without a clear physiological explanation for iron deficiency (especially men and postmenopausal women) should be evaluated by gastroscopy/colonoscopy to exclude a source of GI bleeding, particularly a malignant lesion. Determine possible causes based on history and examination; initiate iron therapy; screen for coeliac disease; discuss timing of scopes with a gastroenterologist.
	Consider thalassaemia if MCH or MCV is low and not explained by iron deficiency, or if long standing. Consider blood loss or haemolysis if reticulocyte count is increased. Seek Haematology advice or, in presence of chronic kidney disease (eGFR <45), Nephrology advice.

3.2.3	Iron Deficiency Anaemia Therapy
	Non-Major Surgery • Prescribe: **Ferrous Fumarate 210mg po bd for 2 days, then tds for 5 weeks** • Provide patient information material & Send GP follow up request.
	Major Surgery for Benign Conditions • Delay surgery for at least 6 weeks. • Prescribe: **Ferrous Fumarate 210mg po bd for 2 days, then tds for 5 weeks** • Provide patient information material & Send GP Information letter. • Repeat FBC after 4 weeks. • If intolerant of oral iron, arrange iv iron therapy.
	Major Surgery for Malignant Conditions or unable to delay & planned in 2 to 6 weeks • Arrange iv iron therapy (see 3.2.4) • Provide patient information material & Send GP Information letter. • Repeat FBC in 3 weeks or on day of admission.
	Major Surgery for Malignant Conditions within 2 weeks • Arrange IV Iron therapy (see 3.2.4) • If Hb <100g/L, arrange attendance on Tudno ward for Blood Transfusion at least 2 days before surgery. When prescribing pRBC units, assume Haemoglobin will have increased by 1g/dl with iv iron. • Provide patient information material & Send GP Information letter. • Repeat FBC on day of admission.

3.2.4	**Intravenous Iron Therapy (Ferinject: Ferric Carboxymaltose)**
	Indication: - Intolerant of oral iron - Surgery within 6 weeks
	Avoid in the first trimester of pregnancy.
	Expected effect on Hb: A rapid Hb response is exhibited in most patients (50% at 5 days, 75% at 10–14 days, maximal at 3 weeks).
	Stop any oral iron prescription
	Side Effects: • Allergic reaction (<1%) • Permanent skin staining (Rare) • Headache, dizziness, rash, nausea, abdominal discomfort, muscle cramps, flushing (1-10%)
	Arranging visit for administration: **Tudno ward, Ysbyty Gwynedd: Ex. 5881** **Ambulatory Care Unit: Ex 8095** Liase with both patient & ward to confirm a convenient time and date. Out of hours service may be offered upon request. In patients with iron deficiency anaemia & eGFR <45, refer to CKD team: Ex. 8469
	Prescribing: **Ferinject 1000mg in 250ml 0.9% saline over 15 minutes** Do not exceed 20mg/kg (maximum 1000mg) in a single dose.
	Administration: (see Appendix A) Dilute dose (maximum 1000mg per dose) in 250ml 0.9% saline Administer by iv infusion over 15 minutes Monitor vital observations and for signs of allergic reaction during and for 30 minutes post infusion.
	Post Administration: Repeat FBC in 2-3 weeks after administration.

3.2.5	**Non-Iron Deficiency Anaemia**
	For folate and B12 deficiency, refer to GP to commence supplements Consider clinical context Review renal function, MCV/MCH and requested tests Request GP Review +/- specialist advice
	For known CKD patients with eGFR <45, refer to CKD Team For patients with eGFR <45 who have no previous history of CKD, make a formal referral to Nephrology.

3.2.6	**Use of HemoCue**
	Capillary blood haemoglobin of all patients undergoing major surgery should be performed in the nurse led POAC using a HemoCue instrument. This will enable anaemia blood investigations to be sent during within the same visit, if appropriate. (See Appendix B)

3.2.7	**Patient information**
	Patients should be provided with the following information: - rationale for blood investigations and recommended treatment. - treatment side effects. - written instructions regarding attendance for treatment and follow up.

3.2.8	**Audit**
	Data will be collected on all patients found to be anaemic at POAC assessment (Appendix D)

4. References

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9. Australian National Blood Authority 2011, Patient Blood Management Guidelines. (www.blood.gov.auh)
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Consultation has taken place with:

Name	Title	Date Consulted
Dr M Jibani	Consultant Nephrologist	February 2015

5. Appendices

Appendix A: Nurse IV Ferrinject administration guide

BCUHB POAC –WEST

FERINJECT® 50mg/mL - Administration Advice

ENSURE ANAPHYLAXIS KIT AND EMERGENCY TROLLEY ARE AT HAND AND IN DATE BEFORE INFUSION

Ferinject® is available in three different size vials:

Volume	Iron content
2mL	100mg
10mL	500mg
20mL	1000mg

Inspect all vials for sediment and damage before use. Only use if it appears to be a sediment-free, homogenous solution. Each Vial is for single use only.

Administration equipment required

Apron and Gloves	Blue Butterflies	Sterile Saline 10ml flush	Mediswabs
Anaphylactic Kit	Blue and Green Venflons and Extension Tubing	Sodium Chloride IV Infusion 250ml and 100ml	Gauze Squares
Tympanic thermometer			
Dynamap	Plastic IV tray	Sharps bin	Micropore
IV Infusion Pump and Drip Stand	Giving set	Tourniquet	Small elastoplast dressing

Infusion preparation

Ferinject® must only be diluted with sterile Sodium Chloride 0.9% solution.

Dilution plan of Ferinject® for intravenous infusion.

Ferinject® (50mg/mL)	Iron	Maximum Volume of sterile Sodium Chloride 0.9% solution	minimum administration time
4-10mL	200-500mg	100mL	6 minutes
10-20mL	500-1000mg	250mL	15 minutes.

Maximum single dose in one week = 1000mg (max 20mg/kg body weight)

Dilutions to concentrations less than 2mg iron/ml are not allowed.

Observation

- Check blood pressure, pulse and temperature (vital observatories) prior to infusion, 5 minutes after commencement and at the end of infusion.
- Check for signs of cannula displacement following commencement of the infusion and periodically through the duration of the infusion. Cannula displacement can cause tissue irritation resulting in pain, local irritation and brown discolouration of the skin. If local irritation occurs stop infusion immediately and inform patient of potential prolonged discolouration of skin.
- Patient must remain on the ward for 30 minutes after the end of the infusion. Vital observation and signs of allergic reaction should be monitored for this period of time.
- Can cause hypersensitivity reaction including anaphylactic reactions.

SHOULD ADVERSE REATIONS BE NOTED AT ANY TIME THROUGHOUT THE INFUSION:

1. ***STOP the infusion immediately.***
2. ***Contact medical assistance.***
3. ***Contact Resuscitation team in the event of cardiac arrest, anaphylactic shock or respiratory distress.***

Side effects

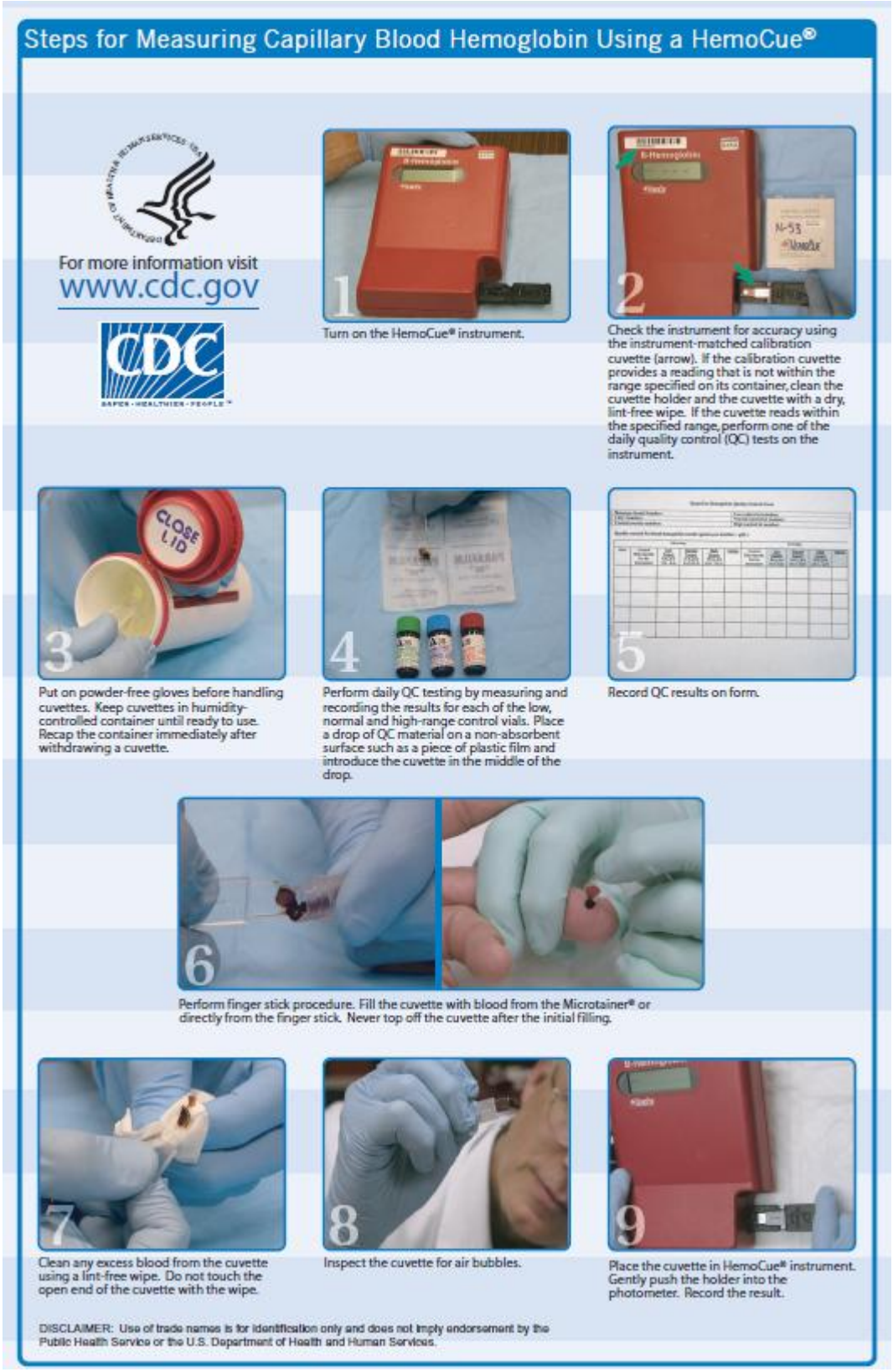
Common side effects include:

- Headache
- Hypotension, hypertension and facial flushing
- Rash
- Injection site reactions
- Nausea, abdominal pain, constipation and diarrhoea.
- Dizziness, light headedness or confusion (advise patients not to drive post infusion until symptoms have ceased if these symptoms occur)

Produced by: Elena Roberts, Lead POAC pharmacist Date Produced: May 2015

Reference: Summary of product characteristics: Ferinject. Last updated 23/10/13. EMC. http://www.medicines.org.uk/emc/medicine/24167 Accessed 05/05/15

Appendix B: HemoCue: Guidance For Use



Appendix C: BCU Operative Procedures Blood Transfusion Requirements

Key	Description
G	Gynae & Obstetrics
O	Orthopaedics
S	General Surgery
U	Urology
V	Vascular Surgery
G/S	Group & Save sample recommended
XM	Cross Match recommended (no. of units stated)
nil	No samples required for G/S or XM

Procedure	Blood	Speciality
Colposuspension	G/S	G
D & C	nil	G
Delorme Procedure	G/S	G
External Cephalic Version	G/S	G
Hysterectomy, Total Abdominal (TAH)	G/S	G
Hysterectomy, Vaginal	G/S	G
Hysterectomy, Wertheimers	XM 4	G
Oophrectomy	G/S	G
Ovarian Cyst	G/S	G
Pelvic Floor Repair (Anterior/Posterior)	G/S	G
Reversal of sterilization	G/S	G
Ankle Fusion	nil	O
Discectomy	G/S	O
Dynamic Hip Screw	G/S	O
Elbow Replacement	G/S	O
Hemiarthroplasty	G/S	O
Removal of Intramedullary Nail	G/S	O
Spinal Fusion	G/S	O
Total Hip Replacement (THR)	G/S	O
THR, Revision	XM 4	O
THR, Bilateral	G/S	O
Total Knee Replacement (TKR)	G/S	O
TKR, Revision	G/S	O
TKR, Bilateral	G/S	O
Total shoulder replacement	nil	O
Zadek procedure	nil	O
A P Resection	XM 2	S
Amputation	nil	S
Amputation, Above Knee	G/S	S
Ampution, Below Knee	XM 2	S
(Lower) Anterior Resection rectum	XM 2	S
Biopsy, Breast	nil	S
Biopsy, Lymph Node	nil	S
Biopsy, Renal	G/S	S
Block dissection neck	XM 2	S
Bronchoscopy	nil	S

Cholecystectomy	G/S	S
Choledocho-jejunostomy	G/S	S
Cleft Palate repair	G/S	S
Colectomy	XM 1	S
Colonoscopy	G/S	S
Colostomy	G/S	S
Endoscopy	G/S	S
ERCP (GI Endoscopy)	G/S	S
Excision Thyroglossal Cyst	G/S	S
Fundoplication (Abd)	G/S	S
Gastrectomy	XM 2	S
Gastrectomy, Partial	XM 2	S
Gastrectomy, Total	XM 4	S
Gastroentrostomy	G/S	S
Gastro-jejunostomy	G/S	S
Hartmann's Procedure	XM 2	S
Hemicolectomy	G/S	S
Hernia Repair	nil	S
Hernia Repair, Incisional	G/S	S
Hernia Repair, Transthoracic	XM 6	S
Ileostomy	G/S	S
Ileostomy, reversal	G/S	S
Insertion Tenchkoff Catheter	G/S	S
Ivor Lewis Oesophagectomy	XM 3	S
Laparoscopy	G/S	S
Laparotomy	XM 2	S
Mastectomy	G/S	S
Mastectomy, Radical	XM 2	S
Oesophagectomy	XM 2	S
OG Duodenoscopy	nil	S
Panproctocolectomy	XM 2	S
Pelvic Clearance	XM 5	S
Pouch procedure	XM 2	S
Proctectomy	XM 1	S
Proctocolectomy	XM 2	S
Sigmoid Colectomy	G/S	S
Small Bowel Resection	G/S	S
Splenectomy	G/S	S
Thyroidectomy, Partial	G/S	S
Thyroidectomy, Sub-total	G/S	S
Thyroidectomy, Total	G/S	S
Tonsillectomy	nil	S
Tracheostomy	XM 2	S
Vagotomy	G/S	S
Varicose veins	nil	S
Whipples Pancreatectomy	XM 6	
Wide local excision	G/S	S
Cystectomy	XM 3	U
Cystoscopy	G/S	U
Nephrectomy	G/S	U
Nephrolithotomy	XM 1	U

PCNL	G/S	U
Prostatectomy, Open	G/S	U
Prostatectomy, Radical	XM 2	U
Prostatectomy, TUR	G/S	U
Pyeloplasty	G/S	U
TUR Bladder Tumour	G/S	U
Abdominal Aneurysm, Elective Repair	XM 4	V
A-V Fistula	G/S	V
Aneurysm, Femoral	XM 3	V
Angiography	G/S	V
Axillo-bifomoral graft	XM 2	V
By-pass, Aorto-Bifemoral	XM 2	V
By-pass, Aortomesenteric	XM 4	V
By-pass, Femoral-Distal	G/S	V
By-pass, Femoral-popliteal	G/S	V
By-pass, Femoro-Tibial	G/S	V
By-pass, Ileo-femoral	G/S	V
Cardiac Catheter	nil	V
Carotid Endarterectomy	G/S	V
Cervical Sympathectomy-Endoscopic	G/S	V
Embolectomy	XM 1	V
Femoro-femoral X-over graft	G&S	V

Appendix D: Patient Data Proforma

Preoperative Haemoglobin Assessment and Optimisation for Elective Major Surgery Proforma

To be completed for ALL patients with Anaemia (Hb <130g/L) undergoing surgery requiring G&S or Cross Match (see BCU Transfusion requirements document)

Refer to Anaemia Assessment & Optimisation Flow Chart

Patient ID

Date of POAC: __/__/__

Planned Procedure:

Operation Date: _____ Not Confirmed ☐

Haemacue: ☐ g/dl

Investigations: Date of Tests :__/__/__

Hb: ☐ g/L Ferritin: ☐ mcg/dl Transferrin Sats: ☐ % CRP: ☐

B12: ☐ Folate: ☐ eGFR: ☐

Evaluation:

Iron Deficiency anaemia: ☐ B12/Folate deficiency: ☐ Other cause for anaemia: ☐

Does this patient require specialist review (UGI, Haematology, Renal)? Y ☐ N ☐

Management:

Category	Tick	Therapy	Date of Therapy Commenced
Can be delayed for >6 weeks		Trial of Ferous Fumarate	
Can be delayed for >6 weeks, but oral iron intolerant		IV Ferinject	
Can be delayed for >2 weeks, but within <6 weeks (e.g. cancer)		IV Ferinject	
Cannot be delayed: within 2 weeks		IV Ferinject & pRBC Transfusion	
Cannot be delayed: within 2 weeks & refuses blood products		IV Ferinject & Erythropoietin	
B12/Folate deficiency		B12/Folate	
Known CKD patients with eGFR <45,		Refer to CKD team	

Outcome after therapy:

Date of test:__/__/__ Repeat Hb: ☐ g/L Ferritin: ☐ mcg/dl Transferrin Sats: ☐ %

Confirmed Operation Date: __/__/__

Place copy in Anaemia folder

Appendix E: Patient Oral Iron Information Letter

GIG CYMRU NHS WALES
Bwrdd Iechyd Prifysgol Betsi Cadwaladr University Health Board
Ysbyty Gwynedd, Bangor

ANAESTHETICS, CRITICAL CARE & PAIN MANAGEMENT CPG

PRE-OPERATIVE ASSESSMENT CLINIC (POAC)

Lead Consultant: Dr C Eickmann Lead Nurse: Anwen Creegan
Pharmacist: Elena Gwyn Roberts (BCUHB - Anaesthetics)I ☎ 01248 384366

Patient identification sticker

Dear patient

We tested your blood at the Pre-Operative Assessment Clinic and it showed that the iron levels were low. This is known as iron-deficiency anaemia.

In order to help improve this deficiency before your surgery, it is recommended that you are started on a course of oral iron tablets. Improving the level or iron in your body may reduce problems during and after your surgery.

A prescription for the following item will have been prepared for you: Please follow the details below for administration

Oral iron tablets:

Dosage:

Frequency:

Duration:

Common side-effects include nausea, stomach upset, constipation and diarrhoea.

Iron supplements are best absorbed when they are taken an hour before food. However, to reduce stomach upset, you may prefer to take your doses after meals.

If you can not tolerate your Iron tablet it is important you contact the Pre-Operative Assessment Clinic on the number above.

If you require more information please do not hesitate to contact us.

Appendix F: Patient IV Iron Information Leaflet

GIG CYMRU NHS WALES | Bwrdd Iechyd Prifysgol Betsi Cadwaladr University Health Board

Ysbyty Gwynedd, Bangor

ANAESTHETICS, CRITICAL CARE & PAIN MANAGEMENT CPG

PRE-OPERATIVE ASSESSMENT CLINIC (POAC)

Lead Consultant: Dr C Eickmann Lead Nurse: Anwen Creegan
Pharmacist: Elena Gwyn Roberts (BCUHB - Anaesthetics)l ☎ 01248 384366

Taflen wybodaeth therapi haearn i mewn i wythïen (Mewnwythiennol)

Wedi i ni brofi eich gwaed yn y Clinig Asesu Cyn Llawdriniaeth (POAC), dangosodd bod y lefelau haearn yn eich gwaed yn isel. Gelwir hyn yn anaemia. Mae eich meddyg wedi penderfynu rhagnodi triniaeth haearn (Ferinject), sef pigiad uniongyrchol i wythïen yn eich braich.

Mae'n bosibl y bydd ar rai cleifion angen dau bigiad gyda bwlch o wythnos rhwng y ddau.

Mae haearn isel yn gallu arwain at flinder a bod yn fyr eich gwynt.
Mae'n bwysig bod gennych ddigon o haearn cyn eich llawdriniaeth i helpu eich corff wella ar ei hôl ac i osgoi unrhyw effeithiau annymunol yn ystod y llawdriniaeth.

Byddwn yn ail brofi eich gwaed 2-3 wythnos ar ôl i chi gael y pigiad haearn i wneud yn siŵr bod y lefelau wedi codi.

Os byddwch yn cymryd tabledi haearn drwy'r geg, dylech roi'r gorau i'w cymryd cyn eich pigiadau Ferinject ac am o leiaf 5 diwrnod ar ôl y pigiad olaf.

Bydd yr apwyntiad i gael y pigiad yn para tua 1 awr.

Ystyrir Ferinject yn ddiogel i'w ddefnyddio. Yn anaml (mewn llai nag 1% o achosion) gall achosi adwaith alergol. Fe gewch eich monitro'n agos o'r herwydd, yn ystod ac wedi'r pigiad.

Cymhlethdod prin ond arwyddocaol o Ferinject ydy staenio neu ddrygliwiad parhaol ar y croen, sy'n gallu digwydd os ydy rhywfaint o'r cyffur yn gollwng tu allan i'r wythïen yn ystod y pigiad. Er mwyn lleihau'r risg, fe roddir dŵr neu doddiant halwynog (dŵr halen) yn eich gwythïen cyn y Ferinject.

Mae sgîl-effeithiau posib yn digwydd mewn rhwng 1 a 10% o gleifion, gan gynnwys cur pen, pendro, brech, cyfogi a chwydu, poen abdomenol, gwasgfa yn y cyhyrau, dolur rhydd, rhwymedd, gweithgarwch anarferol yr isu/sfu, gwrido, pwysedd gwaed isel neu uchel ac adweithiau ar leoliad pigiadau.

Iron therapy into a vein (Intravenous) information leaflet.

We tested your blood in the Pre-Operative Assessment Clinic (POAC) and it showed that the iron level in your blood was low. This is known as anemia. Your doctor has decided to prescribe iron treatment (Ferinject). This will be an injection you will receive directly into a vein in your arm.

Some patients may need two injections with a gap of at least one week between them.

Low iron can give you symptoms of feeling tired and breathless.
It is important to have enough iron before your surgery to help your body recover afterwards and to avoid any unwanted effects during your surgery.

We will re-test your bloods 2-3 three weeks after you receive the iron injections to make sure your levels have risen.

If you are taking oral iron tablets you should stop them before your Ferinject injections and for at least 5 days after the last injection.

The appointment for the injection will last about one hour.

Ferinject is considered safe to use. Rarely (in under 1% of cases) it can cause allergic reactions. You will therefore be monitored closely before, during and after the infusion.

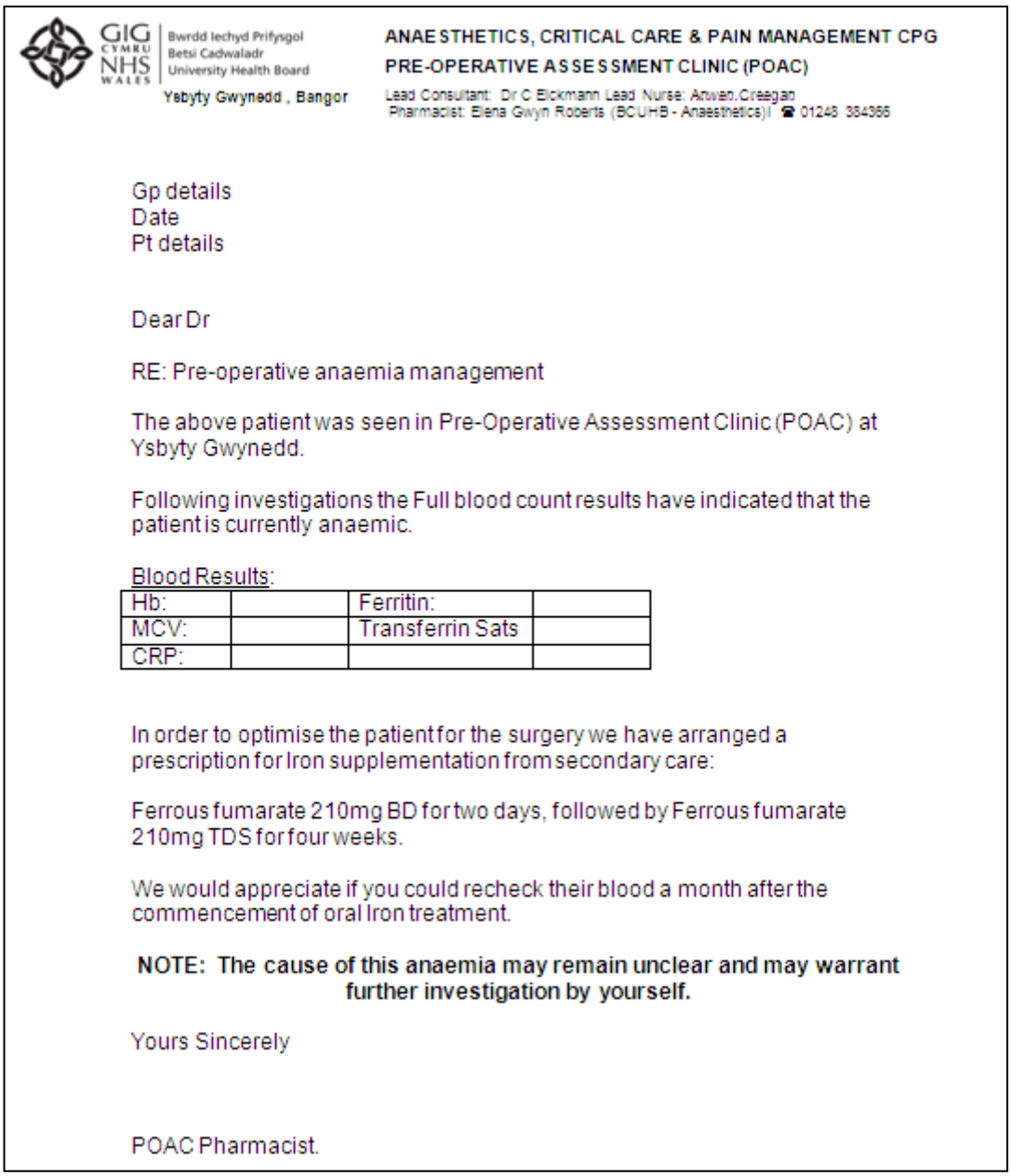
A rare but significant complication of Ferinject is permanent skin staining or discoloration, which can occur if some of the drug leaks outside the vein during the infusion. To reduce the risk, a flush of water or saline (salt solution) is given into your vein before the Ferinject.

Potential mild side effects occur in 1 to 10% of patients, including headache, dizziness, rash, nausea and vomiting, abdominal pain, muscle cramps, diarrhoea, constipation, abnormal liver function, flushing, low or high blood pressure and injection site reactions.

This information is available in other languages or formats upon request.
Please contact us on the above number if required.

Review date: September 2020
Version number: 2
Author/Job title: Elena Roberts/Lead POAC Pharmacist.

Appendix G: GP Letter: Oral Iron



GIG CYMRU NHS WALES
Bwrdd Iechyd Prifysgol Betsi Cadwaladr University Health Board
Ysbyty Gwynedd , Bangor

ANAESTHETICS, CRITICAL CARE & PAIN MANAGEMENT CPG
PRE-OPERATIVE ASSESSMENT CLINIC (POAC)

Lead Consultant: Dr C Eickmann Lead Nurse: Arwen.Creegan
Pharmacist: Elena Gwyn Roberts (BCUHB - Anaesthetics)| ☎ 01248 384366

Gp details
Date
Pt details

Dear Dr

RE: Pre-operative anaemia management

The above patient was seen in Pre-Operative Assessment Clinic (POAC) at Ysbyty Gwynedd.

Following investigations the Full blood count results have indicated that the patient is currently anaemic.

Blood Results:

Hb:		Ferritin:	
MCV:		Transferrin Sats	
CRP:			

In order to optimise the patient for the surgery we have arranged a prescription for Iron supplementation from secondary care:

Ferrous fumarate 210mg BD for two days, followed by Ferrous fumarate 210mg TDS for four weeks.

We would appreciate if you could recheck their blood a month after the commencement of oral Iron treatment.

NOTE: The cause of this anaemia may remain unclear and may warrant further investigation by yourself.

Yours Sincerely

POAC Pharmacist.

Appendix H: GP Letter: IV Iron

GIG CYMRU NHS WALES
Bwrdd Iechyd Prifysgol Betsi Cadwaladr University Health Board
Ysbyty Gwynedd, Bangor

ANAESTHETICS, CRITICAL CARE & PAIN MANAGEMENT CPG
PRE-OPERATIVE ASSESSMENT CLINIC (POAC)
Lead Consultant: Dr C Eickmann Lead Nurse: Anwen Creegan
Pharmacist: Elena Gwyn Roberts (BCUHB - Anaesthetics) ☎ 01248 384366

Date
Pt details

Dear Dr

RE: Pre-operative anaemia management

The above patient was seen in Pre-Operative Assessment Clinic (POAC) at Ysbyty Gwynedd.

Following investigations the Full blood count results have indicated that the patient is currently anaemic.

Blood Results:

Hb:		Ferritin:	
MCV:		Transferrin Sats	
CRP:			

In order to optimise the patient for the surgery it was decided that the patient would receive intravenous Iron therapy.

Infusion date	Dose of Ferinject®	Comments

Patient has been advised to stop any oral iron supplement at initiation of IV iron and for at least 5 days after the **last** infusion.

We will recheck the bloods 2-3 weeks post infusion.

NOTE: The cause of this anaemia may remain unclear and may warrant further investigation by yourself.

Yours Sincerely

POAC Nurse.